Class 10: Structural Bioinformatics P1

Kai Christman (A17522126)

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Background

The main repistory for high-resolution structural data on biomolecules is called the **Protein Data Bank (PDB)**.

PDB Statistics

What is in the PDB in terms of molecule type and structure determination method?

Read a CSV file of current PDB stats obtained from: https://www.rcsb.org/stats/summary

```
pdb <- read.csv("data export summary.csv")
pdb
```

```
           Molecular.Type   X.ray     EM    NMR Integrative Multiple.methods
1          Protein (only) 180,758 23,111 12,813         348              229
2 Protein/Oligosaccharide  10,488  3,741     34           8               11
3              Protein/NA   9,205  6,751    287          26                8
4     Nucleic acid (only)   3,154    250  1,578           3               15
5                   Other     178     27     35           4                0
6  Oligosaccharide (only)      11      0      6           0                1
  Neutron Other   Total
1      84    32 217,375
2       1     0  14,283
```

```
3       0     0  16,277
4       3     1   5,004
5       0     0     244
6       0     4      22
```

Seems like the data for some of these mesuarements is in Characters rather than integers, due to the presence of a comma! We can fix this with the following code, utilizing the `gsub()` and the `as.numeric()` functions:

```
cols_to_fix <- c("X.ray", "EM", "NMR")

rm.comma <- pdb[cols_to_fix] <- lapply(pdb[cols_to_fix], function(x) {
  as.numeric(gsub(",", "", x))
})

pdb
```

```
           Molecular.Type  X.ray    EM   NMR Integrative Multiple.methods
1          Protein (only) 180758 23111 12813         348              229
2 Protein/Oligosaccharide  10488  3741    34           8               11
3              Protein/NA   9205  6751   287          26                8
4     Nucleic acid (only)   3154   250  1578           3               15
5                   Other    178    27    35           4                0
6  Oligosaccharide (only)     11     0     6           0                1
  Neutron Other   Total
1      84    32 217,375
2       1     0  14,283
3       0     0  16,277
4       3     1   5,004
5       0     0     244
6       0     4      22
```

We could also use a different import function: `read_csv` from the **readr** package. This CSV importer speaks "American" (i.e. deals with commas in numbers in a comma seperated value file).

```
library(readr)

pdb_data <- read_csv("data export summary.csv")
```

```
Rows: 6 Columns: 9
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): Molecular Type
dbl (4): Integrative, Multiple methods, Neutron, Other
num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
pdb_data
```

```
# A tibble: 6 x 9
  `Molecular Type`     `X-ray`    EM   NMR Integrative `Multiple methods` Neutron
  <chr>                  <dbl> <dbl> <dbl>       <dbl>              <dbl>   <dbl>
1 Protein (only)        180758 23111 12813         348                229      84
2 Protein/Oligosacch~    10488  3741    34           8                 11       1
3 Protein/NA              9205  6751   287          26                  8       0
4 Nucleic acid (only)     3154   250  1578           3                 15       3
5 Other                    178    27    35           4                  0       0
6 Oligosaccharide (o~       11     0     6           0                  1       0
# i 2 more variables: Other <dbl>, Total <dbl>
```

> Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
n.tot <- sum(pdb_data$Total)
n.xray <- sum(pdb_data$`X-ray`)
n.EM <- sum(pdb_data$EM)

xray_pct <- n.xray / n.tot
EM_pct <- n.EM / n.tot

xray_pct
```

```
[1] 0.8048577
```

```
EM_pct
```

```
[1] 0.1338046
```

Q2: What proportion of structures in the PDB are protein?

```
total_protein <- pdb_data$Total[1]

(total_protein / 202556314) * 100
```

```
[1] 0.1073158
```

The total number of protein sequences is: 202,556,314. (This is sourced from UniProt) > **Key-point**: We have a very, very small strucutural coverage of known proteins (~0.1%). Most structures we know about (~80%) come from one method (X-ray crsytalography).

SKIPPED THIS QUESTION IN LAB > Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

Visualizing PDB data with Mol-star

Main stand alone web version with all features is at: https://molstar.org/viewer/

Figure 1: The HIV-protease enzyme is a homodimer with two chains

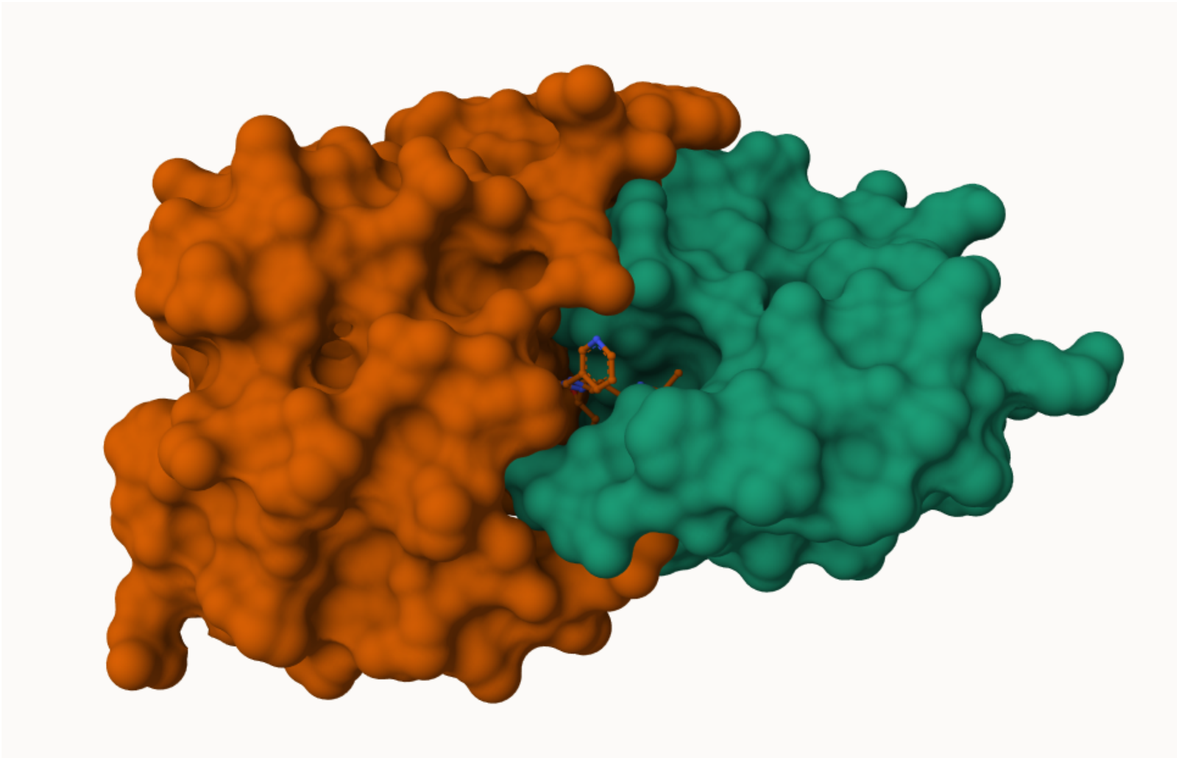

Figure 2: The HIV-Protease enzyme with Molecular surface Representation

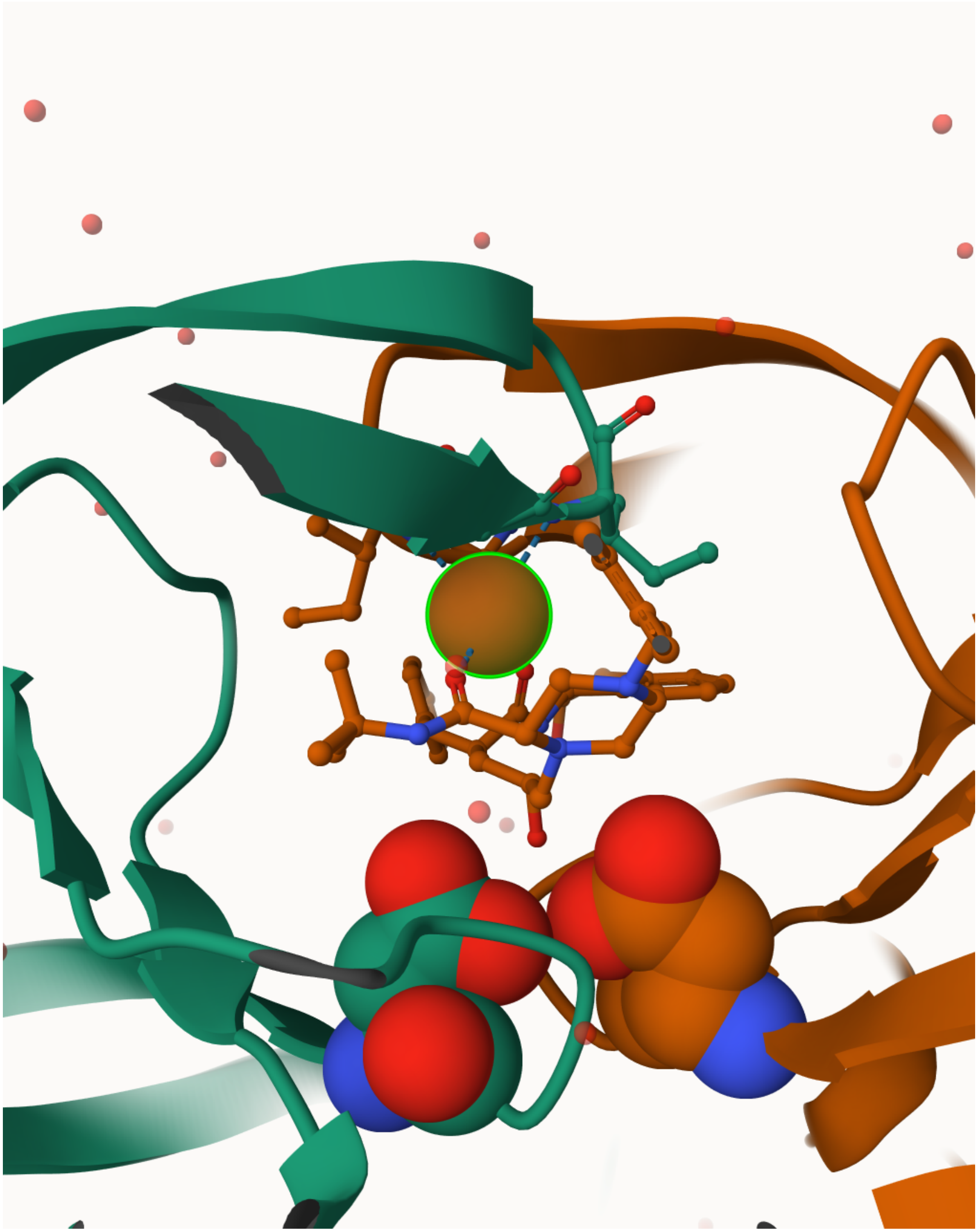

Figure 3: Spacefill display of catalytic ASP25 amino acids and key binding site water molecule